Barath R

Embedded Systems & Firmware Engineer | RISC-V, RTOS & IoT Systems

Chennai, India | +91-9442169681 | barath5727@gmail.com
github.com/barath0508 | linkedin.com/in/barath0508 | barath0508-portfolio.vercel.app

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Date: October 16, 2026
To: Hiring Team / Talent Acquisition
Company: **[Target Company Name]**
Subject: Application for **[Embedded Systems / Firmware Engineer Intern]**

Dear Hiring Team,

I am writing to express my enthusiastic interest in the **[Position Title]** opportunity at **[Company Name]**. As a third-year Electronics and Communication Engineering student at Rajalakshmi Institute of Technology (CGPA: 8.65/10), I specialize in bare-metal systems programming, real-time operating systems (**FreeRTOS** and **Zephyr RTOS**), and hardware bring-up. My technical journey has centered on developing close to the silicon—translating architecture datasheets and register maps into deterministic firmware, designing multi-layer PCBs, and building reliable telemetry pipelines.

Key highlights of my technical background include:

- **Bare-Metal RISC-V Architecture:** Architected a bare-metal trap handler and preemptive context switcher in RISC-V assembly and C with 0-byte RTOS overhead on the ESP32-C6 MCU. By directly configuring Memory-Mapped I/O (MMIO), machine timer interrupts (MTIE), and trap vectors (mtvec), I achieved < 2.4 µs context-switch latency across 32 registers.
- **Real-Time Flight Firmware & RTOS:** Engineered real-time flight control firmware for an autonomous quadcopter (ESP-Drone) in C/C++ on ESP-IDF and FreeRTOS, sustaining a 500 Hz closed-loop attitude stabilization cycle (< 1% loop jitter), 6-DOF IMU sensor fusion with quaternion estimation, and low-latency bidirectional telemetry via ESP-NOW (< 5 ms).
- **Hardware Bring-up & PCB Design:** Designed and routed 2- and 4-layer PCBs using KiCad and Proteus, followed by SMD soldering and bench testing with oscilloscopes, logic analyzers, multimeters, and JTAG/SWD debuggers. Implemented analog sensor bridges (differential LDR sensing) and simulated conformal microstrip patch antenna arrays in CST Studio.
- **Technical Leadership & Mentorship:** As Executive Technical Organizer for the IETE Students' Forum (ISF), mentored 500+ student engineers across firmware development, sensor interfacing, and microcontroller bring-up while driving 10+ embedded and IoT builds end-to-end.

What attracts me most to **[Company Name]** is your engineering standard in building mission-critical, high-reliability embedded hardware. I am eager to apply my foundation in Embedded C/C++, RTOS concurrency, hardware bring-up, and low-level debugging to solve tangible engineering challenges on your team.

Thank you for your time and consideration. I would welcome the opportunity to discuss how my firmware and hardware skill set aligns with your engineering goals.

Sincerely,

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